Econometrics 120A: Discussion Section

Bivariate Distributions and Sampling

Anita Sammarini & Linghui Wu

Department of Economics

Joint, marginal and conditional distributions

Recap: joint and marginal distributions

S = n° of H in two tosses
Y = whether the 2nd toss is heads or tails

- **Joint distribution:** $P[X = x,\ Y = y]$ for every pair (x, y).
- **Valid** if ❶ every entry ≥ 0 and ❷ all entries sum to 1.
- **Marginals:** sum out the other variable

$$P[X = x] = \sum_{\text{all } y} P[X = x, Y = y]$$

$$P[Y = y] = \sum_{\text{all } x} P[X = x, Y = y]$$

- Joint $\Rightarrow$ marginals: always possible. Marginals $\Rightarrow$ joint: only under independence.

$S \backslash Y$	0	1	$P(s)$
0	1/4	0	**1/4**
1	1/4	1/4	**1/2**
2	0	1/4	**1/4**
$P(y)$	**1/2**	**1/2**	1

Lecture Example 2: two coin tosses, S = number of heads, Y = second toss.

Row sums $\rightarrow$ marginal of S.
Column sums $\downarrow$ marginal of Y.

Outcome	S	Y
TT	0	0
TH	1	1
HT	1	0
HH	2	1

- $P(S=1, Y=1) = P(TH) = \frac{1}{4}$ JOINT PROBABILITY because we are imposing two conditions simultaneously

- WHAT IF I ONLY CARE ABOUT THE NUMBER OF HEADS AND DON'T CARE ABOUT THE SECOND TOSS?
  ↳ we add the possibilities that correspond to the same number of heads

$$P(S=1) = P(S=1, Y=0) + P(S=1, Y=1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$ MARGINAL!

CONDITIONING → means restricting our attention to a smaller population and recalculating probabilities within that population

$S=1 \Rightarrow \{TH, HT\}$

$$P(Y=0 \mid S=1) = \frac{1}{2}$$

$$P(Y=0 \mid S=1) = \frac{P(Y=0, S=1)}{P(S=1)} = \frac{1/4}{1/2} = \frac{1}{2}$$

↳ divide by the conditioning event because that is our new universe

INDEPENDENCE → $P(X=x \mid Y=y) = P(X=x)$

Conditional Distribution Questions

Problem Set 3, Question 6

The joint distribution of two random variables X and Y is given by

$X \backslash Y$	0	1
0	0.1	0
1	0.4	0.4
2	0	0.1

(a) What is the conditional distribution of X given that $Y = 1$?

(b) What is the mean of X given $Y = 1$?

(c) What is the variance of X given $Y = 1$?

(d) Suppose we are trying to learn the mean of X, but observe X only when $Y = 1$ (e.g. X = number of holidays a person takes per year, Y = whether their phone number is listed, so they can be included in the survey). Otherwise the survey is well run. What is the bias in our estimate of the mean of X?

Problem Set 3, Question 6

The joint distribution of two random variables X and Y is given by

$X \backslash Y$	0	1
0	0.1	0
1	0.4	0.4
2	0	0.1

(a) What is the conditional distribution of X given that $Y = 1$?

$$P(X|Y=1) = \frac{P(X=x, Y=1)}{P(Y=1)}$$

$$P(Y=1) = \sum_x P(X=x, Y=1) = 0 + 0.4 + 0.1 = 0.5$$

$$P(X=0|Y=1) = \frac{P(X=0, Y=1)}{P(Y=1)} = \frac{0}{0.5} = 0$$

$$P(X=1|Y=1) = \frac{P(X=1, Y=1)}{P(Y=1)} = \frac{0.4}{0.5} = 0.8$$

$$P(X=2|Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{0.1}{0.5} = 0.2$$

$X\|Y=1$	1
0	0
1	0.8
2	0.2

Problem Set 3, Question 6

The joint distribution of two random variables X and Y is given by

$X \backslash Y$	0	1
0	0.1	0
1	0.4	0.4
2	0	0.1

(b) What is the mean of X given $Y = 1$?

$$E(X|Y=1) = \sum_x x \, P(X|Y=1) =$$

$$= 0 \cdot 0 + 1 \cdot 0.8 + 2 \cdot 0.2 = 1.2$$

Problem Set 3, Question 6

The joint distribution of two random variables X and Y is given by

$X \backslash Y$	0	1
0	0.1	0
1	0.4	0.4
2	0	0.1

(c) What is the variance of X given $Y = 1$?

$$\text{Var}(X|Y=1) = \sum_x (x - E(X|Y=1))^2 \cdot P(X|Y=1) =$$

$$= (0-1.2)^2 \cdot 0 + (1-1.2)^2 \cdot 0.8 + (2-1.2)^2 \cdot 0.2 = 0.16$$

Problem Set 3, Question 6

The joint distribution of two random variables X and Y is given by

$X \backslash Y$	0	1
0	0.1	0
1	0.4	0.4
2	0	0.1

(d) We observe X only when $Y = 1$ (only people with a listed number get surveyed). What is the bias in our estimate of the mean of X?

The unconditional distribution is:

$P(X=0) = 0.1$
$P(X=1) = 0.8$
$P(X=2) = 0.1$

$$E(X) = \sum_x P(X=x) = 0(0.1) + 1(0.8) + 2(0.1) = 1$$

$$\text{Bias} = E(X|Y=1) - E(X) = 1.2 - 1 = \underline{\underline{0.2}}$$

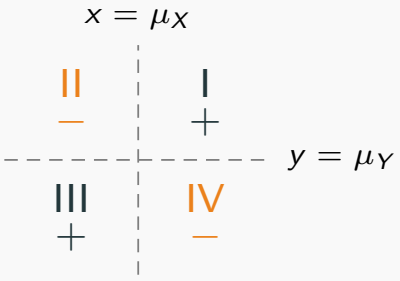
Covariance and correlation

Recap: from sample to population

- DS4, sample: $s_{xy} = \frac{1}{n-1}\sum_i (x_i - \bar{x})(y_i - \bar{y})$
- Population: weight each pair by its probability

$$\mathrm{Cov}(X, Y) = \sum_x \sum_y (x - \mu_X)(y - \mu_Y) \times P[X = x, Y = y]$$

- Shortcut: $\mathrm{Cov}(X, Y) = E[XY] - E[X]\,E[Y]$
- Correlation: $\rho_{XY} = \dfrac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$, unitless, $-1 \le \rho \le 1$
- Independent $\Rightarrow$ Cov $= 0$.
  Cov $= 0$ $\not\Rightarrow$ independent: covariance only picks up *linear* co-movement.



Same quadrant logic as DS4, but each (x, y) now counts with weight $P[X = x, Y = y]$.

Lecture Example 2:
$\mathrm{Cov}(S, Y) = \frac{1}{4}$,

$$\rho = \frac{1/4}{\sqrt{1/2}\sqrt{1/4}} \approx 0.71$$

$X \perp Y \Rightarrow \mathrm{cov}(X,Y) = 0$ BUT $\mathrm{cov}(X,Y) = 0 \not\Rightarrow X \perp Y$

Problem Set 3, Question 4

Consider the following joint distribution of X and Y.

$X \backslash Y$	0	1
1	0.1	0.2
2	0.2	0.2
3	0.1	0.2

(a) Compute the mean of X and the mean of Y.

(b) Compute the covariance of X and Y.

(c) Without doing additional math, what does this result suggest regarding the relationship between X and Y? Are they related in any way?

(d) What is the mean of $g(X, Y) = Y/X$?

Problem Set 3, Question 4

Consider the following joint distribution of X and Y.

$X \backslash Y$	0	1
1	0.1	0.2
2	0.2	0.2
3	0.1	0.2

(a) Compute the mean of X and the mean of Y.

MARGINALS →

$$P(X=1) = \sum_y P(X=1, Y=y) = 0.1 + 0.2 = 0.3$$

$$P(X=2) = \sum_y P(X=2, Y=y) = 0.2 + 0.2 = 0.4$$

$$P(X=3) = \sum_y P(X=3, Y=y) = 0.1 + 0.2 = 0.3$$

$$P(Y=0) = \sum_x P(X=x, Y=0) = 0.1 + 0.2 + 0.1 = 0.4$$

$$P(Y=1) = \sum_x P(X=x, Y=1) = 0.2 + 0.2 + 0.2 = 0.6$$

$$E(X) = \sum_x x\, P(X=x) = 0(0.3) + 2(0.4) + 3(0.3) = 2$$

$$E(Y) = \sum_y y\, P(Y=y) = 0(0.4) + 1(0.6) = 0.6$$

USEFUL PROPERTY → $E(Y) = P(Y=1)$ IF $Y \in \{0,1\}$

Problem Set 3, Question 4

Consider the following joint distribution of X and Y.

$X \backslash Y$	0	1
1	0.1	0.2
2	0.2	0.2
3	0.1	0.2

(b) Compute the covariance of X and Y.

$\mu_X = 2$

$\mu_Y = 0.6$

$$Cov(X,Y) = \sum_x \sum_y (x-\mu_x)(y-\mu_y) P(X=x, Y=y)$$

$$= (1-\mu_x)(0-\mu_y) P(X=1,Y=0) + (1-\mu_x)(1-\mu_y) P(X=1,Y=1) +$$
$$+ (2-\mu_x)(0-\mu_y) P(X=2,Y=0) + (2-\mu_x)(1-\mu_y) P(X=2,Y=1) +$$
$$+ (3-\mu_x)(0-\mu_y) P(X=3,Y=0) + (3-\mu_x)(1-\mu_y) P(X=3,Y=1) =$$

$$= (-1)(-0.6)(0.1) + (-1)(0.4)(0.2) + 0 + 0 + 1(-0.6)(0.1)$$
$$+ (1)(0.4)(0.2) =$$

$$= 0.06 - 0.08 - 0.06 + 0.08 = 0$$

$$E(XY) = 1(0.2) + 2(0.2) + 3(0.2) = 1.2$$

$$Cov(X,Y) = E(XY) - E(X)\cdot E(Y) = 1.2 - 2(0.6) = 0$$

Problem Set 3, Question 4

Consider the following joint distribution of X and Y.

$X \backslash Y$	0	1
1	0.1	0.2
2	0.2	0.2
3	0.1	0.2

(c) Without doing additional math, what does this result suggest about the relationship between X and Y? Are they related in any way?

$\text{Cov}(X,Y) = 0 \not\Rightarrow X \perp Y$

check independence $\rightarrow$ $P(X=1, Y=0) = 0.1$

$P(X=1)\,P(Y=0) = (0.3)(0.4) = 0.12$

$\Rightarrow P(X=1, Y=0) \neq P(X=1)\,P(Y=0)$

NOT INDEPENDENT

Problem Set 3, Question 4

Consider the following joint distribution of X and Y.

$X \backslash Y$	0	1
1	0.1	0.2
2	0.2	0.2
3	0.1	0.2

(d) What is the mean of $g(X, Y) = Y/X$?

$R = \frac{Y}{X}$ new random variable!

$r = y/x$	$P(R=r)$
0	$0.1+0.2+0.1 = 0.4$
$1/3$	0.2
$1/2$	0.2
1	0.2

$$E(R) = 0\cdot(0.4) + 0.2\,(1/3 + 1/2 + 1) = \frac{11}{30} \approx 0.37$$

Note: $E(Y/X) \neq \frac{E(Y)}{E(X)}$

Functions of two random variables

Recap: means and variances of $g(X, Y)$

- New random variable $R = g(X, Y)$. Two ways to get $E[R]$:
  1. Find the distribution of R, then $E[R] = \sum_r r\, P[R = r]$
  2. Directly: $E[g(X, Y)] = \sum_x \sum_y g(x, y)\, P[X = x, Y = y]$

  Same answer; pick whichever is easier.
- In general $E[g(X, Y)] \neq g(E[X], E[Y])$. Linear functions are the exception: ↳ $E(x^2) \neq E(x)^2$

Mean	$E[aX + bY] = aE[X] + bE[Y]$
Variance	$\mathrm{Var}(aX + bY) = a^2 \mathrm{Var}(X) + b^2 \mathrm{Var}(Y) + 2ab\, \mathrm{Cov}(X, Y)$

- The mean rule always holds, independent or not.
- If X, Y are independent the highlighted term is 0.
- With $a = b = \frac{1}{2}$ we get the average of two random variables: the key step for sampling.

Problem Set 3, Question 5

Suppose we have the joint distribution between two random variables X and Y such that

$X \backslash Y$	0	1
-1	$3/8$	0
0	$1/8$	$3/8$
1	0	$1/8$

(a) We want to create a new random variable $X + Y$. Compute the mean of $X + Y$.

(b) Compute the variance of $X + Y$.

Problem Set 3, Question 5

Suppose we have the joint distribution between two random variables X and Y such that

$X \backslash Y$	0	1
-1	3/8	0
0	1/8	3/8
1	0	1/8

(a) Compute the mean of the new random variable $X + Y$.

$R = X+Y$	$P(R)$
-1	3/8
0	1/8
1	3/8
2	1/8

$$E(R) = -1\left(\tfrac{3}{8}\right) + 0\left(\tfrac{1}{8}\right) + 1\left(\tfrac{3}{8}\right) + 2\left(\tfrac{1}{8}\right) = \tfrac{1}{4}$$

or

$$E(X+Y) = E(X) + E(Y) = -\tfrac{1}{4} + \tfrac{1}{2} = \tfrac{1}{4}$$

$$E(X) = -1 \cdot \tfrac{3}{8} + \tfrac{1}{8} = -\tfrac{1}{4}$$

$$E(Y) = \tfrac{3}{8} + \tfrac{1}{8} = \tfrac{1}{2}$$

Problem Set 3, Question 5

Suppose we have the joint distribution between two random variables X and Y such that

$X \backslash Y$	0	1
-1	3/8	0
0	1/8	3/8
1	0	1/8

(b) Compute the variance of $X + Y$.

$R = X+Y$	$P(R)$
-1	3/8
0	1/8
1	3/8
2	1/8

$$E(R^2) = \frac{3}{8} + \frac{3}{8} + 4 \cdot \frac{1}{8} = \frac{10}{8}$$

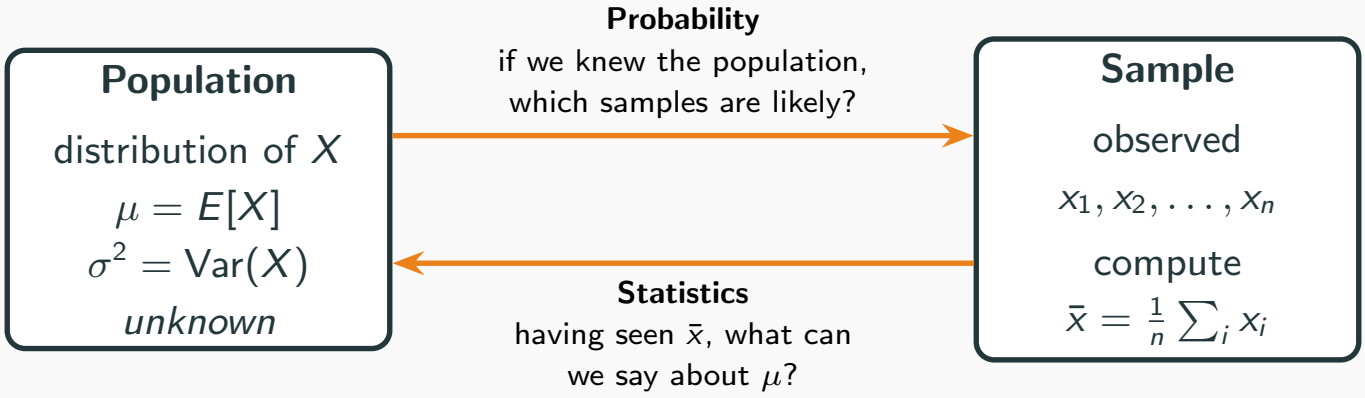
$$Var(R) = E(R^2) - E(R)^2 = \frac{10}{8} - \left(\frac{1}{4}\right)^2 = \frac{19}{16}$$

$$Var(X+Y) = Var(X) + Var(Y) + 2\,Cov(X,Y)$$

check!

Sampling

The big picture



- ⇨ The bridge between the two: the distribution of $\bar{X}$ (the **sampling distribution**).
- ⇨ It tells us which sample averages are likely if our model of the population is right, so we can check the model against the data.

A sample is n random variables

- **Before** we collect data, each observation is a random variable: $X_1, X_2, \ldots, X_n$.
  After, we see numbers: $x_1, x_2, \ldots, x_n$.
- Poll example: $X_i = 1$ if person i supports the candidate, 0 otherwise, so $X_i \sim \text{Bernoulli}(p)$.
- **Very Simple Random Sample (VSRS):**
  1. **Independent:** $P[X_1 = x_1, X_2 = x_2] = P[X_1 = x_1]\, P[X_2 = x_2]$
  2. **Identically distributed:** every X_i has the same distribution, so the same μ and σ^2
- So a sample is just a big joint distribution. Everything from the first half of today applies, with n variables instead of two.

KNOWING THE FIRST OBSERVATION DOESN'T CHANGE THE PROBABILITY DISTRIBUTION OF THE SECOND OBSERVATION

EXAMPLE A. I FLIP TWO COINS, ONE HAS 50% CHANCE OF HEADS AND THE OTHER HAS A 90% CHANCE. => INDEP BUT NOT IDENTICALLY DISTRIBUTED

EXAMPLE B. TWO STUDENTS STUDY TOGETHER, WEEKLY HOURS OF STUDY HAVE SAME MARGINAL DISTRIBUTION BUT REALIZATIONS CORRELATED => IDENTICALLY DISTRIBUTED BUT DEPENDENT

Moments of the Sample Mean

Using the linear combination rules

$$E(\bar{X}) = E\left(\frac{1}{m}\sum_{i=1}^{m} X_i\right) = \frac{1}{m}\sum_{i=1}^{m} E(X_i) = \frac{1}{m}\cdot m\mu$$

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad \text{(a linear combination with weights } 1/n\text{)}$$

Mean $\quad E[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n}\, n\mu = \mu$ (linearity only)

Variance $\quad \mathrm{Var}(\bar{X}) = \frac{1}{n^2}\Big[\underbrace{\textstyle\sum_i \mathrm{Var}(X_i)}_{n\sigma^2 \text{ (identical)}} + \underbrace{\textstyle\sum_{i\neq j} \mathrm{Cov}(X_i, X_j)}_{0 \text{ (independent)}}\Big] = \frac{\sigma^2}{n}$

- $n = 2$: $\mathrm{Var}\left(\frac{1}{2}X_1 + \frac{1}{2}X_2\right) = \frac{1}{4}\sigma^2 + \frac{1}{4}\sigma^2 + 2\cdot\frac{1}{4}\cdot 0 = \frac{\sigma^2}{2}$ (the $\mathrm{Var}(aX + bY)$ rule with $a = b = \frac{1}{2}$).
- **Warning:** σ = spread of *one* observation; $\sigma/\sqrt{n}$ = spread of the *average*. Ask yourself: is the question about one person, or about an average?

The Shape of $\bar{X}$

Three results

sums & averages of independent normal r.v. remain normal.

1. **Normal data, exact:** if $X_i \sim N(\mu, \sigma^2)$ then $\bar{X} \sim N(\mu, \sigma^2/n)$ for any n.
2. **Law of Large Numbers:** for any $\varepsilon > 0$, $P\left[|\bar{X}_n - \mu| > \varepsilon\right] \to 0$ as $n \to \infty$.
   $\mathrm{Var}(\bar{X}) = \sigma^2/n \to 0$, so $\bar{X}$ piles up on μ. This is why we average.
3. **Central Limit Theorem:** if $X_i \sim (\mu, \sigma^2)$, *any* distribution, then for large n

$$\bar{X}_n \overset{a}{\sim} N\left(\mu, \frac{\sigma^2}{n}\right)$$

Recipe (same normal table as DS4):

$$P[\bar{X} > c] \approx P\left[Z > \frac{c - \mu}{\sigma/\sqrt{n}}\right], \qquad Z \sim N(0,1)$$

The Central Limit Theorem

CLT for Bernoulli random variables

$X_i \sim \text{Bernoulli}(p)$: $\mu = p$, $\sigma^2 = p(1-p)$ $\Rightarrow$

$$\hat{p} = \bar{X} \overset{a}{\sim} N\left(p, \frac{p(1-p)}{n}\right)$$

$X_i \sim$ Bernoulli (p)

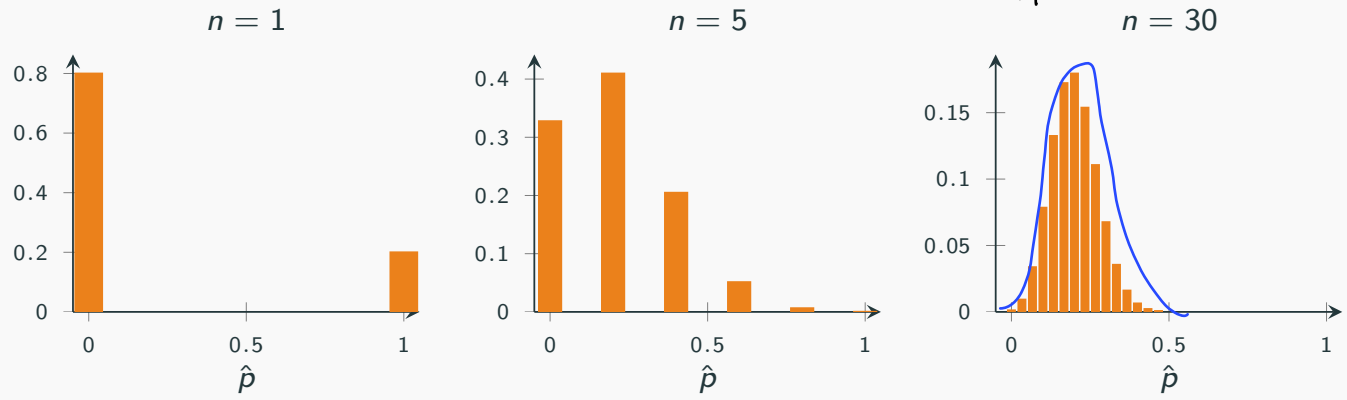
$E(X_i) = p$

$\text{Var}(X_i) = p(1-p)$

$\hat{p} = \frac{1}{m}\sum_i X_i = \bar{X}$

$E[\hat{p}] = p$

$\text{Var}(\hat{p}) = p(1-p)/m$



Exact distribution of $\hat{p} = S/n$ with $S \sim \text{Binomial}(n, 0.2)$. Skewed at $n = 1$; by $n = 30$ it is bell shaped and concentrated near $p = 0.2$.

Lecture example: $n = 400$, $p = 0.6$: $\sigma_{\hat{p}} = \sqrt{0.6 \cdot 0.4/400} \approx 0.0245$.

Sampling Exercise

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

(a) What are $E[\bar{X}]$ and the standard deviation of $\bar{X}$?

(b) Hours are not normally distributed. Why can we still use the normal distribution for $\bar{X}$? What is its approximate distribution?

(c) Approximate the probability that the sample average is above 14 hours.

(d) A classmate concludes: "So only about 2% of students study more than 14 hours a week." What is wrong with this?

(e) Redo (c) with $n = 144$. Which result from lecture does the change illustrate?

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

(a) What are $E[\bar{X}]$ and the standard deviation of $\bar{X}$?

$$E(\bar{X}) = \mu = 12$$

$$\sigma(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{6}{\sqrt{36}} = 1$$

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

(b) Hours are not normally distributed. Why can we still use the normal distribution for $\bar{X}$? What is its approximate distribution?

CLT: X_i are a random sample with finite variance and $n = 36$ is reasonably large so the skewness of individual hours washes out in the average

$$\bar{X} \sim N\left(12, 1\right)$$

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

(c) Approximate the probability that the sample average is above 14 hours.

z	0.00	0.01	0.02	0.03	0.04
1.9	0.0287	0.0281	0.0274	0.0268	0.0262
2.0	0.0228	0.0222	0.0217	0.0212	0.0207
2.1	0.0179	0.0174	0.0170	0.0166	0.0162

Table: $P(Z > z)$

$$P(\bar{X} > 14) \approx P\left(Z > \frac{14 - 12}{6/\sqrt{36}}\right) = P(Z > 2) = 0.0228$$

Sampling Exercise

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

(d) A classmate concludes: "So only about 2% of students study more than 14 hours a week." What is wrong with this?

Mixes up σ and $\sigma/\sqrt{n}$: 2.28% is about the average of 36 students

One student has σ = sd = 6 so 14 hours is only $\frac{14-12}{6} = \frac{1}{3}$ sd above the mean : more than 2% of students study over 14 hours

Individual hours are skewed $\Rightarrow$ the normal table does not apply to one student : CLT is about $\overline{X}$, not X_i

Exercise

The number of hours UCSD students study per week has mean $\mu = 12$ and standard deviation $\sigma = 6$. The distribution is *skewed to the right* (a few students study a lot). You survey a random sample of $n = 36$ students and compute the sample mean $\bar{X}$.

(e) Redo (c) with $n = 144$. Which result from lecture does the change illustrate?

$$\sigma(\bar{X}) = 6/\sqrt{144} = 0.5$$

$$P(\bar{X} > 14) \approx P\left(Z > \frac{14-12}{0.5}\right) = P(Z > 4) \approx 0$$

Quadrupling n halves the spread of $\bar{X}$ $\Rightarrow$ piles up closer to μ.

Summary

- **Joint $\to$ marginal:** sum out the other variable. **Conditional:** slice and rescale.
- **Independence** must hold in *every* cell; equivalently, every conditional equals the marginal.
- **Covariance** $= 0$ does not mean independent; a constant conditional mean does not either.
- $E[aX + bY] = aE[X] + bE[Y]$ always; $\text{Var}(aX + bY)$ needs the $2ab\,\text{Cov}$ term.
- A random sample is n independent, identically distributed random variables, so $E[\bar{X}] = \mu$, $\text{Var}(\bar{X}) = \sigma^2/n$.
- **LLN:** $\bar{X}$ gets close to μ. **CLT:** $\bar{X} \overset{a}{\sim} N(\mu, \sigma^2/n)$ whatever the distribution of X_i.